Beating the Pool, Not the Game

A White-Box Equilibrium Agent for Language-Based Economic Games

Sudhanshu Srivastava · Faisal Aqlan University of Louisville

Interpreting Agent Behavior (IAB)
Workshop at NeurIPS 2026
GLEE Competition Track

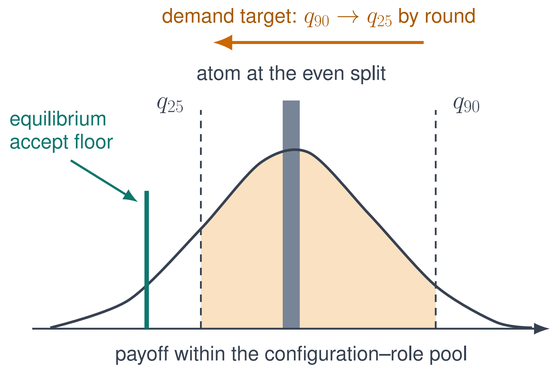
Each game is scored as a payoff percentile against prior play on the same configuration, so our rule-based agent demands what the pool concedes and accepts what equilibrium allows. It makes **zero inference-time LLM calls**, and a replay reproduces **139,071 of 139,072 moves**.

1 Setting: the game is the pool

The GLEE Competition, built on the GLEE benchmark [1], pits agents against a mixed pool of humans, LLM-driven agents, and organizer baselines in three families of two-player games.

- In **bargaining**, players alternate proposals to split a pot M, and player i's value decays by δ_i per round.
- In **negotiation**, a seller and a buyer, with reservation values V_s and V_b on a known grid, exchange price offers.
- In **persuasion**, a seller sees each product's quality over 20 rounds and messages a buyer, who decides whether to buy at price c.

The scoring is relative. Each payoff becomes a percentile among all payoffs on the same configuration and role, adjusted for opponent strength, and the game rating is $2000 + 8000\,(\mathrm{pct} - 0.5)$. Our ablations use the same scale against the public corpus alone.



The opponent to outperform is the empirical distribution of play, which concentrates near even splits, fair prices, and truthful recommendation [1]. **Equilibrium decides what to accept. The pool decides what to demand.**

2 Agent design: clamp to the pool

Our bargaining proposer demands at most

$$k_t = \min\Big(\max\big(\min(k_t^{\mathrm{SPE}},\ 0.70M),\ k_t^{\mathrm{pool}},\ 0.52M\big),\ 0.80M\Big)$$

- k_t^{SPE} is the share backward induction assigns the proposer: the Ståhl recursion [2] on known horizons and the asymmetric Rubinstein share [3] otherwise.
- k_t^{pool} is the round-indexed pool target: the q_{90} payoff early, then down to q_{25}, corrected for our own decay. Where the quantile tables miss a configuration, the band $[0.52M, 0.70M]$ applies alone.
- Offers that clear a $0.9\,\delta_m$ haircut on the continuation share are accepted, offers at or above the pool's q_{90} are never refused, and a patient hand ($\delta_m \geq 0.97$) refuses early offers below the pool median.
- The agent never walks away in bargaining, since rejection weakly dominates.

Negotiation

We anchor at our end of the valuation grid, concede along a Boulware curve [4] toward the opponent's reservation estimate, and in round $T-1$ counter at the opponent's value: the final-counter gambit.

Persuasion

Where uninformed buying pays ($pv > c$), the seller recommends every product. Otherwise it lies only within a trust budget, a capped fraction of the Bayesian persuasion rate $q^* = \min\{p(v-c)/((1-p)c),\ 1\}$ [5]. The buyer keeps a beta-binomial posterior on seller accuracy and stops after one verified bad purchase from a text-mode seller.

- One stateless function per family
- Typed moves with guaranteed-legal fallbacks
- Median move under a millisecond, zero LLM calls

3 Results

2056
primary agent rating

2098
best agent, same policy

117,280
games at $0 inference cost

26th of 71
ranked accounts

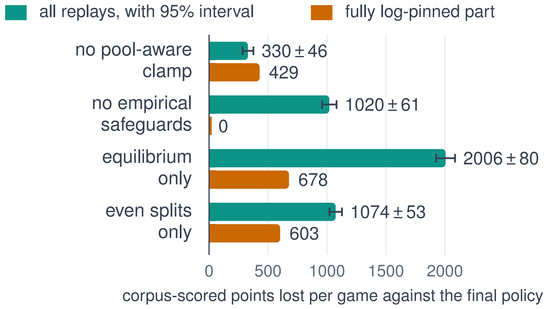
By family, the primary agent rated 2031 in bargaining, 1938 in negotiation, and 2199 in persuasion.

Concurrent live ablation

For 45 hours (Aug 24–26) in one shared pool, arm A, an even-split opener with early q_{25} accepts, lost 188 ± 60 corpus-scored points per game, while the two identical baselines differed by only 19 ± 74 (95% intervals).

Replay on frozen opponent pools

We replay 4,000 logged bargaining games per version on the Aug 28–30 pool, with three threshold draws each.



The clamp is worth 274–330 points on all three frozen pools, and fully log-pinned replays carry it alone. The safeguard gap appears only in modeled play. Even when log-pinned, equilibrium-only and even-split play lose 418–678 and 443–603 points.

4 White-box behavior audit

Every move is a deterministic function of the visible state (persuasion mixing aside), so alignment is checked by replay.

100.0% exact match

139,071 of 139,072 deterministic post-freeze moves reproduced to the cent by replaying the log through the frozen policy (messages excluded).

- 0 strategy exceptions
- 0 crash-loop cooldowns
- 0 bargaining walk-aways
- no message over the platform cap

Behavioral signatures after the freeze

- The final-counter gambit closed 216 of 246 attempts (670 of 738 over the season) at a median 0.99 share of the agreement zone.
- The even-split opener, kept from arm A only in the cells where it won, drew 33% round-1 acceptance with complete and 74% with hidden information.
- Seller lying stayed inside its cap under the trust budget.
- Where the buyer stop-loss governs, no text-mode game had more than one verified bad purchase.

Limitations

The pool keeps moving, so fixed targets go stale. The replay uses the corpus alone and non-reactive opponents, and it roughly doubles arm A's live effect, so its effect sizes may run high. The move logs hold other participants' messages and are not released.

github.com/sudsho/glee-competition-agent
doi.org/10.5281/zenodo.24043059
glee-competition.com
sudhanshu.srivastava@louisville.edu

[1] Shapira et al., GLEE, arXiv:2410.05254, 2024. [2] Ståhl, Bargaining Theory, 1972. [3] Rubinstein, Econometrica, 1982. [4] Faratin, Sierra, and Jennings, Robotics and Autonomous Systems, 1998. [5] Kamenica and Gentzkow, American Economic Review, 2011.